

CLV & Retention Intelligence Report

Customer Lifetime Value & Retention Intelligence Report

Run: 2025-Q4 · Generated 2026-08-31

Data: Online Retail II · Calibration period ending 2011-06-09

Executive Summary

CUSTOMERS ANALYZED	TOTAL PORTFOLIO CLV (6-MO)	OVERALL CHURN RATE
4,942	£3,732,858	54.0%

This report analyzes 4,942 customers with a combined predicted 6-month CLV of £3,732,858. Based on a recency-derived threshold, 54.0% of the base is currently classified as churned. Segmentation converged to 2 cohorts — High-Value / Low-Risk and Low-Value / At-Risk — reflecting a -0.67 correlation between predicted value and churn probability in this customer base, rather than four independently-varying risk/value combinations. The Low-Value / At-Risk cohort carries the highest average churn risk (97.3%) but represents 13.6% of total portfolio value, while the High-Value / Low-Risk cohort — 48.4% of customers — holds 86.4% of portfolio value at comparatively low risk.

Cohort Breakdown

Segmentation converged to **2 cohorts** (K=2, chosen by silhouette score), not a hardcoded four-quadrant structure — driven by a **-0.67** correlation between predicted CLV and churn probability in this customer base (see Model Validation Appendix for the full silhouette/elbow justification).

COHORT	CUSTOMERS	% OF BASE	AVG CLV	AVG CHURN RISK	% OF PORTFOLIO VALUE
High-Value / Low-Risk	2,391	48.4%	£1,349	7.7%	86.4%
Low-Value / At-Risk	2,551	51.6%	£198	97.3%	13.6%

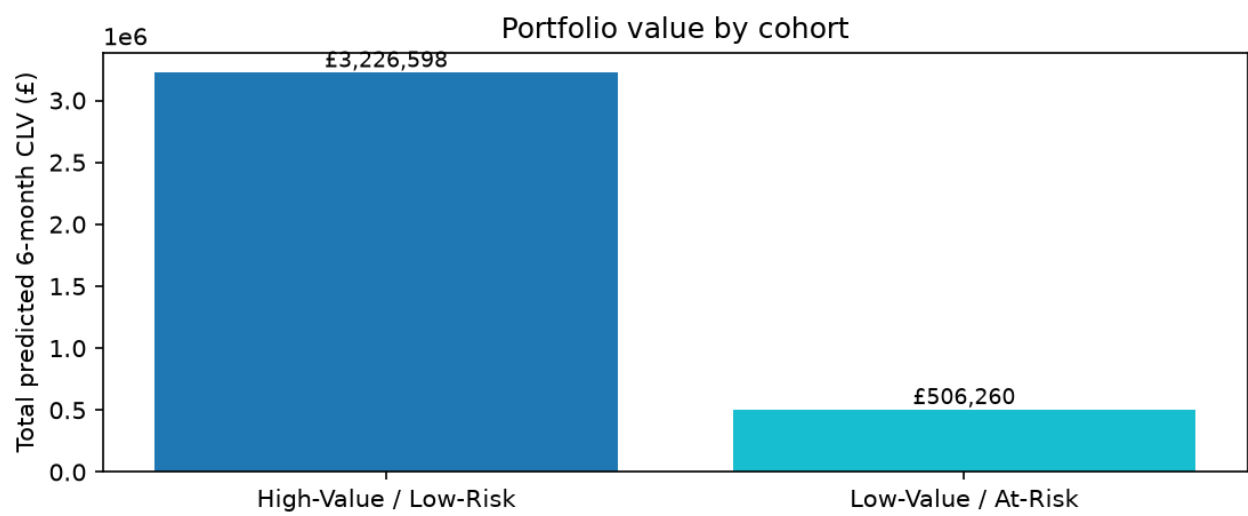


Figure 1. Total predicted 6-month CLV by cohort.

High-Value / Low-Risk — recommended action

Protect and grow. Low churn urgency — prioritize loyalty treatment, cross-sell, and proactive relationship management over reactive retention spend.

Low-Value / At-Risk — recommended action

Scalable, low-cost win-back only. Broad automated re-engagement (e.g. an email offer) may be worthwhile in aggregate, but individually-targeted retention spend is unlikely to be justified by the CLV at stake.

Priority Watchlist

Top 20 customers ranked by **expected value at risk** (predicted CLV × predicted churn probability) — the customers where retention spend has the largest expected payoff, combining both how much they're worth and how likely they are to leave.

CUSTOMER ID	CLV	CHURN PROB.	DAYS SINCE LAST PURCHASE	VALUE AT RISK	RECOMMENDED ACTION
12346	£19,070	96.6%	142	£18,418	Immediate personal outreach
12433	£3,007	98.2%	183	£2,953	Immediate personal outreach
15808	£3,016	97.3%	123	£2,935	Immediate personal outreach
18251	£2,862	98.1%	262	£2,808	Immediate personal outreach
12454	£2,299	98.5%	212	£2,266	Immediate personal outreach
13680	£2,238	97.8%	154	£2,188	Immediate personal outreach
15497	£2,125	98.0%	136	£2,082	Immediate personal outreach
14045	£2,108	98.1%	154	£2,068	Immediate personal outreach
13174	£2,090	96.7%	133	£2,021	Immediate personal outreach
14439	£1,938	98.3%	136	£1,904	Immediate personal outreach
17044	£1,929	96.2%	191	£1,855	Immediate personal outreach
15332	£1,648	98.0%	183	£1,615	Immediate personal outreach
14134	£1,656	96.6%	199	£1,599	Immediate personal outreach
13615	£1,551	97.5%	135	£1,512	Immediate personal outreach
14299	£1,539	97.0%	135	£1,493	Immediate personal outreach
12409	£1,450	98.3%	237	£1,425	Immediate personal outreach
15950	£1,381	97.2%	176	£1,343	Immediate personal outreach

CUSTOMER ID	CLV	CHURN PROB.	DAYS SINCE LAST PURCHASE	VALUE AT RISK	RECOMMENDED ACTION
15523	£1,330	96.0%	205	£1,278	Immediate personal outreach
13734	£1,278	99.5%	325	£1,272	Immediate personal outreach
14175	£1,236	98.3%	146	£1,215	Immediate personal outreach

Model Validation Appendix

CLV Model (BG/NBD + Gamma-Gamma)

METRIC	VALUE
MAE	£614
RMSE	£3,523
MAPE (nonzero actuals)	98.8%
Median APE (nonzero actuals)	47.3%
Spearman rank correlation	0.624

Why MAPE » median APE

Mean MAPE is pulled well above the typical customer's error by a small number of customers whose spend changed sharply between the calibration and holdout periods (e.g. a previously active customer who nearly stopped purchasing). Median APE and Spearman rank correlation are the more representative numbers for a targeting use case, where ranking customers correctly matters more than exact dollar forecasts.

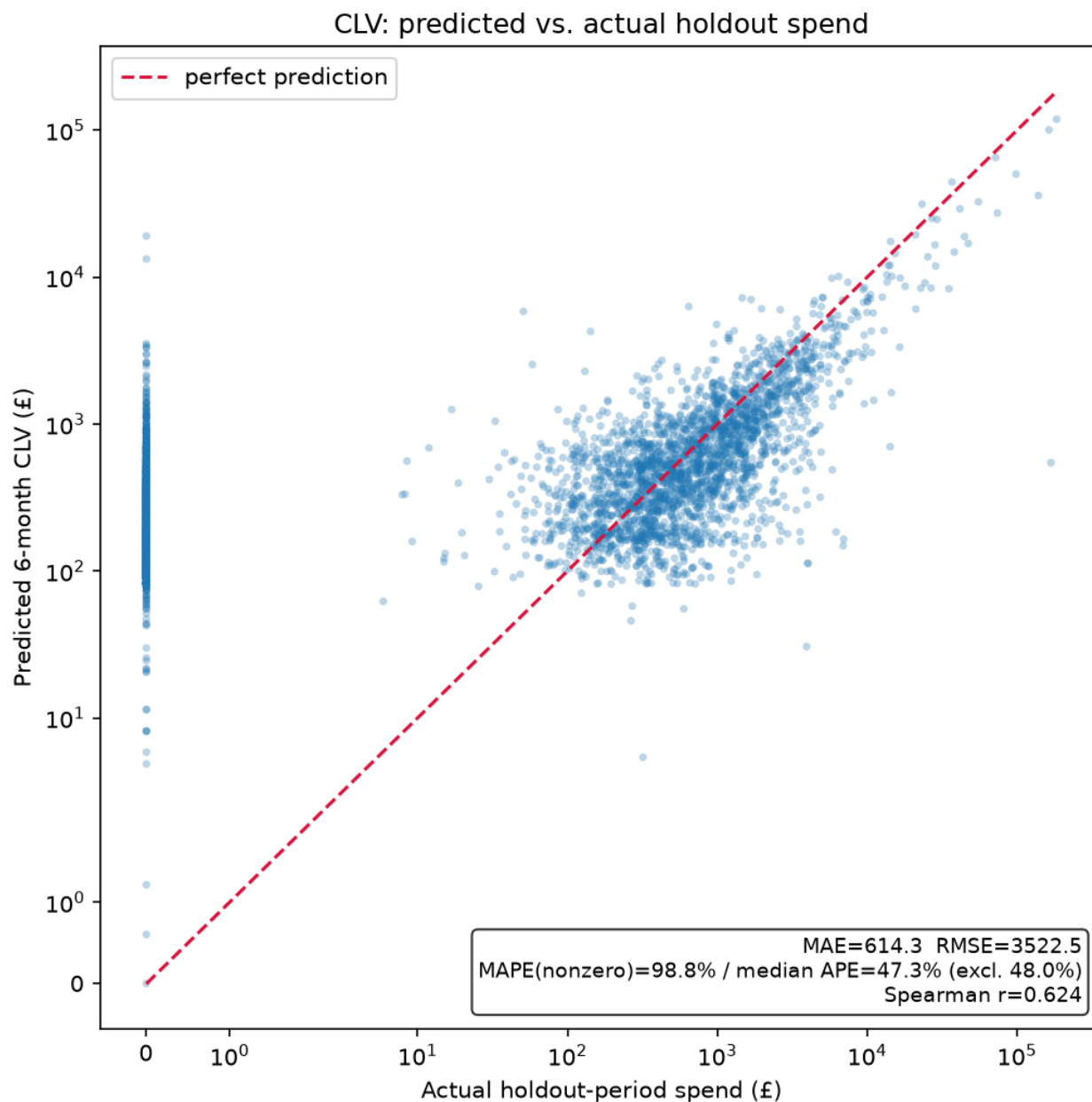


Figure 2. Predicted vs. actual holdout-period spend.

Churn Model (XGBoost)

MODEL	ROC-AUC	INTERPRETATION
Recency-only baseline	1.0	Tautological by construction — the churn label is itself a threshold on recency.
Non-recency features only	0.955	The genuine finding: behavioral features predict churn without seeing recency at all.
Full feature set	1.0	Also ~1.0 — dominated by the same recency feature as the baseline.

SHAP feature importance (full model, mean |SHAP|)

FEATURE	MEAN SHAP
days_since_last_purchase	4.168
purchase_trend	0.345
frequency	0.253
T	0.231
monetary_value	0.205
inter_purchase_interval_std	0.115

Residual-leakage check on the 0.955 non-recency AUC

Before reporting 0.955 as a real result, two secondary leakage paths were checked directly:

- **Tenure (T) mechanical constraint:** days-since-last-purchase $\leq$ T always, so T below the 119-day threshold guarantees "not churned." True for 10.3% of customers (507) — but T's own single-feature AUC is only 0.522, barely above chance.
- **purchase_trend drop test:** this feature correlates with recency by construction (derived from purchase timing). Dropping it entirely: AUC only falls to 0.934 (from 0.955).
- No single non-recency feature's solo AUC exceeds 0.765 (frequency) — the combined result reflects genuine multi-feature signal, not one feature secretly encoding the label.

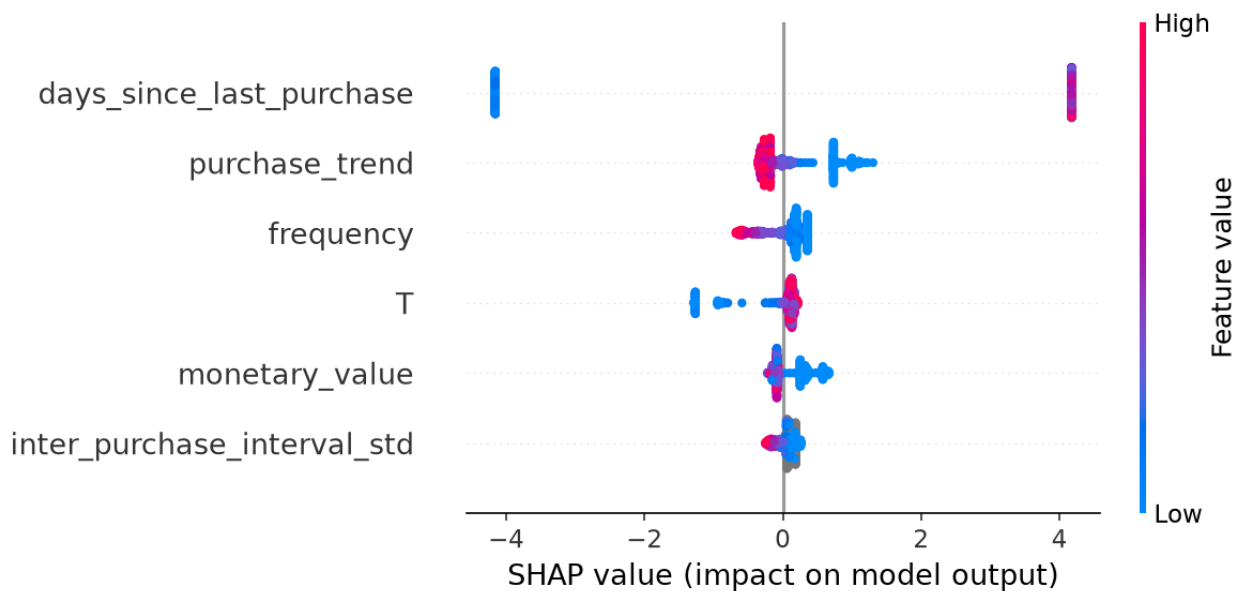


Figure 3. SHAP summary — full churn model.

Segmentation

K=2 chosen by silhouette score (0.462), which decreases monotonically across K=2..8 — no local peak at a higher K. This reflects a real property of the data: log(CLV) and churn probability correlate at -0.67, acting as one dominant axis rather than four independent quadrants.

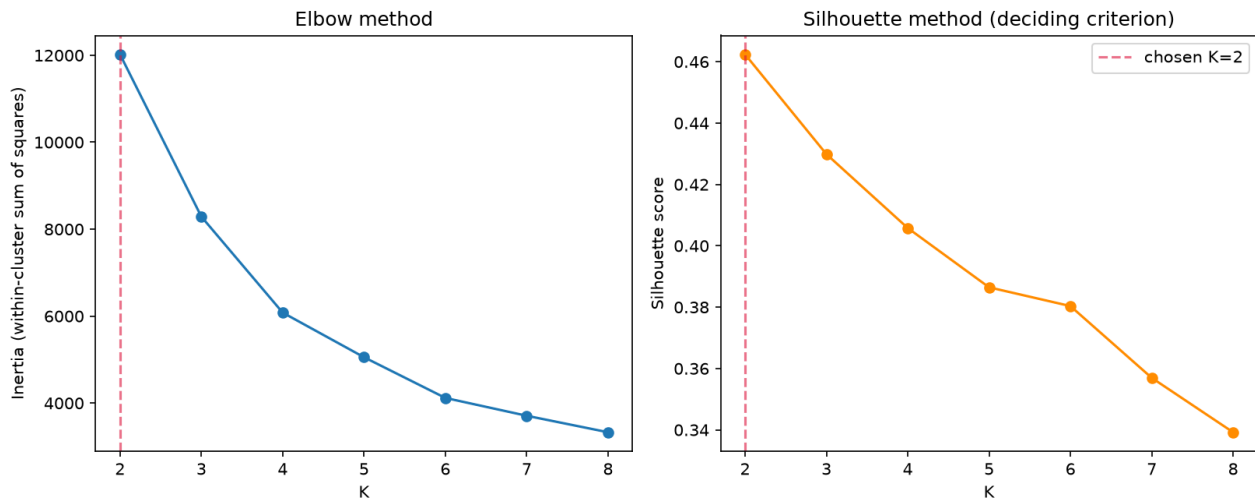


Figure 4. Elbow (inertia) and silhouette score by K.

RFM-only stability check

Adjusted Rand Index = 0.417 between the full-feature clustering and an RFM-only clustering (same K, same customers). Interpreted directly rather than against a pass/fail bar: meaningfully above 0 (chance agreement) but well below 1 (identical clustering) — CLV and churn probability shift roughly half of customers' cluster assignment relative to RFM alone, indicating they add real information rather than only restating RFM.

Bug caught before shipping

Cluster labels were initially derived by comparing each cohort's mean churn probability against the *population median*, mirroring the CLV-labeling logic. Because the churn classifier is so well-separated, the population median churn probability is itself ~0.98 — so a cohort averaging 97.3% churn probability still fell fractionally under that skewed median and was labeled "Low-Risk." Fixed by comparing against a fixed 0.5 threshold instead, since churn probability (unlike CLV) has a real absolute reference point.

Gamma-Gamma Independence Check

Pearson correlation between frequency and monetary value (repeat customers only): $r = 0.129$, under the 0.3 warning threshold — no caveat needed.

Simulated Campaign ROI

Illustrative simulation, not a validated campaign result. No real campaign has been run against this customer base — there is no experimental or historical response data to validate against. This section demonstrates how CLV and churn probability could be used to size and prioritize a retention campaign, under stated assumptions below.

Stated assumptions

- Campaign size: **500** customers contacted (~10.1% of the 4,942-customer base).
- Cost per outreach: **£5.0** (e.g. an email/direct-mail retention offer).
- Assumed retention lift: **15.0%** — a targeted customer's churn probability is assumed to fall by this many percentage points as a result of being contacted, applied uniformly to whoever is targeted (a simplification: real intervention effectiveness would vary by customer, and this lift figure is illustrative, not measured).
- Expected value saved = (targeted customers' total predicted CLV) × retention lift.

STRATEGY	TARGETS	COST	EXPECTED VALUE SAVED	NET ROI	ROI MULTIPLE
Recommended (high-value + high-churn-risk)	500	£2,500	£100,503	£98,003	40.20x
Random targeting (same size)	500	£2,500	£56,650	£54,150	22.66x

Recommended targeting produces an estimated **£43,853** more expected value saved than random targeting of the same size, purely from prioritizing customers where CLV and churn risk are both high — this uplift is a direct consequence of the targeting rule, not of the (unvalidated) retention lift assumption, which is held identical across both strategies.

Data & Methodology Notes

Dataset

Online Retail II (UK online retailer), 2009-12-01 to 2011-12-09. 1,067,371 raw transaction line items → 802,644 after cleaning. Calibration period ends 2011-06-09; churn/segmentation are derived from calibration-period behavior only.

Cleaning summary

STEP	ROWS DROPPED	REASON
drop_null_customer_id	243,007	No customer to attribute the purchase to
drop_cancelled_invoices	18,744	Invoice prefixed 'C' denotes a cancellation, not a completed purchase
drop_non_product_stock_codes	2,913	Postage/discount/adjustment/fee entries are not purchased goods
drop_low_quantity	0	Quantity < 1 is a return/adjustment, not a purchase
drop_low_price	63	Price < 0.01 is a data error or non-sale line item

Limitations

- **Null CustomerID exclusion:** 22.8% of raw rows had no CustomerID and were dropped entirely — those purchases are invisible to this entire analysis.
- **Duplicate rows:** 3.2% (26,055 rows) of cleaned transactions are exact duplicates of unclear origin (repeat entries vs. export artifact) — retained as-is, since deduping would require an assumption the data doesn't support.
- **Single-country skew:** the dataset is predominantly UK transactions; findings won't generalize to other markets without revalidation.
- **Segmentation reflects a 2-axis structure, not four independent quadrants:** log(CLV) and churn probability correlate at -0.67 in this customer base, so value and risk move together strongly enough that K-Means finds one dominant axis of separation rather than a "high/low value × high/low risk" grid. A future run with a more heterogeneous customer base might not show this pattern.

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